DP014	**DP014**
Physics 1	**Fizik 1**
Semester I	**Semester I**
Session 2023/2024	**Sesi 2023/2024**
2 hours	**2 jam**

UNIT FIZIK
KOLEJ MATRIKULASI KELANTAN

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI 3
MATRICULATION PROGRAMME PRE-EXAMINATION 3

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **9** halaman bercetak.
This question paper consists of **9** *printed pages.*

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum *Laju cahaya dalam vakum*	c	$= 3.00 \times 10^{8}$ m s^{-1}
Permeability of free space *Ketelapan ruang bebas*	μ_0	$= 4\pi \times 10^{-7}$ H m^{-1}
Permittivity of free space *Ketelusan ruang bebas*	ε_0	$= 8.85 \times 10^{-12}$ F m^{-1}
Electron charge magnitude *Magnitud cas elektron*	e	$= 1.60 \times 10^{-19}$ C
Planck constant *Pemalar Planck*	h	$= 6.63 \times 10^{-34}$ J s
Electron mass *Jisim elektron*	m_e	$= 9.11 \times 10^{-31}$ kg $= 5.49 \times 10^{-4}$ u
Neutron mass *Jisim neutron*	m_n	$= 1.674 \times 10^{-27}$ kg $= 1.008665$ u
Proton mass *Jisim proton*	m_p	$= 1.672 \times 10^{-27}$ kg $= 1.007277$ u
Deuteron mass *Jisim deuteron*	m_d	$= 3.34 \times 10^{-27}$ kg $= 2.014102$ u
Molar gas constant *Pemalar gas molar*	R	$= 8.31$ J K^{-1} mol^{-1}
Avogadro constant *Pemalar Avogadro*	N_A	$= 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant *Pemalar Boltzmann*	k	$= 1.38 \times 10^{-23}$ J K^{-1}
Free-fall acceleration *Pecutan jatuh bebas*	g	$= 9.81$ m s^{-2}
Atomic mass unit *Unit jisim atom*	1 u	$= 1.66 \times 10^{-27}$ kg $= 931.5\ \frac{MeV}{c^2}$
Electron volt *Elektron volt*	1 eV	$= 1.6 \times 10^{-19}$ J

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Constant of proportionality for Coulomb's law *Pemalar hukum Coulomb*	$k = \frac{1}{4\pi\varepsilon_{\circ}}$	$= 9.0 \times 10^{9}\ \mathrm{N\,m^{2}\,C^{-2}}$
Atmospheric pressure *Tekanan atmosfera*	1 atm	$= 1.013 \times 10^{5}$ Pa
Density of water *Ketumpatan air*	ρ_{w}	= 1000 kg $\mathrm{m^{-3}}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \frac{1}{2}(u + v)t$
5. $v = u - gt$
6. $v^2 = u^2 - 2gs$
7. $s = ut - \frac{1}{2}gt^2$
8. $p = mv$
9. $J = F\Delta t$
10. $J = \Delta p = mv - mu$
11. $f_s \leq \mu_s N$
12. $f_k = \mu_k N$
13. $W = \vec{F} \cdot \vec{s} = Fs\cos\theta$
14. $K = \frac{1}{2}mv^2$
15. $U = mgh$
16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$
18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$
19. $s = r\theta$
20. $v = r\omega$
21. $a_t = r\alpha$
22. $\omega = \omega_\circ + \alpha t$
23. $\theta = \omega_\circ t + \frac{1}{2}\alpha t^2$
24. $\omega^2 = {\omega_\circ}^2 + 2\alpha\theta$
25. $\theta = \frac{1}{2}(\omega_\circ + \omega)t$
26. $\tau = rF\sin\theta$
27. $\frac{Q}{t} = -kA\left(\frac{\Delta \mathrm{T}}{L}\right)$
28. $v_{rms} = \sqrt{< v^2 >}$
29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$
30. $\Delta U = Q - W$

Answer **all** questions.

1. (a) A piece of anthracite has a volume of 15 cm^3 and a mass of 27 g. What is its density in SI units?

[2 *marks*]

(b) Three forces of magnitude 6 N, 2 N and 3 N act on a small object in directions north, south and west respectively. Find the magnitude and direction of the resultant force.

[6 *marks*]

2. (a) A car initially moves at a speed of 20 m s^{-1} and undergoes uniform acceleration at a rate of 2 m s^{-2} for 6 s. It then maintains a constant speed for half a minute. The brakes are applied, causing the car to uniformly decelerate until it comes to a stop in 5 seconds.

(i) Find the maximum speed reached by the car.

(ii) Sketch the speed versus time graph to represent the journey.

(iii) Calculate the total distance covered by the car.

[7 *marks*]

(b) A stone is thrown vertically upwards with an initial velocity 12 m s^{-1}. Neglecting air resistance, find

(i) the maximum height.

(ii) the time taken before it reaches the ground.

[5 *marks*]

3. (a) A trolley having a mass of 570 g moves at a constant velocity of 21 m s^{-1} before colliding with a wall and bouncing back. The wall exerts an impulse of 20.5 N s on the trolley. Calculate

(i) the velocity of the trolley after it hits the wall.

(ii) the magnitude of average force acting on the trolley while in contact with the wall for 0.6 s.

[4 *marks*]

(b)

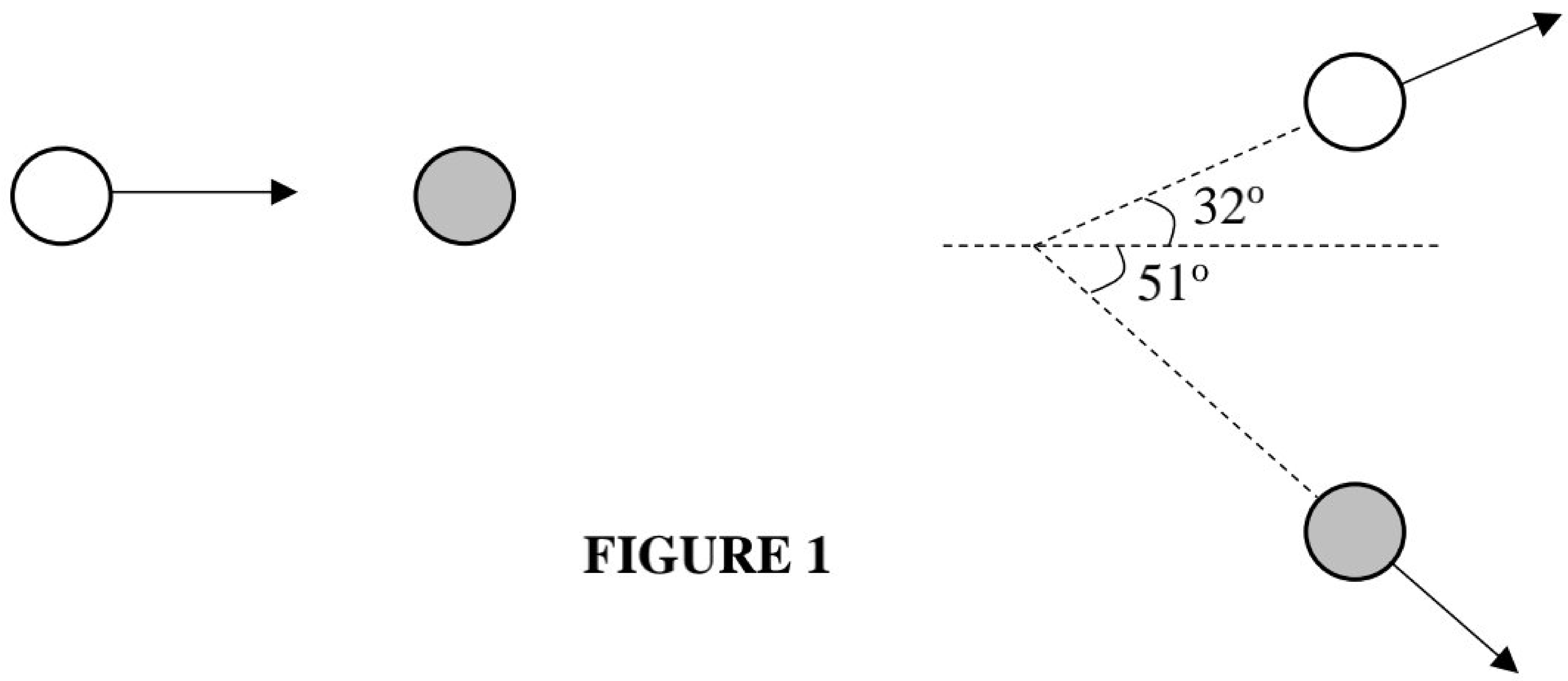

FIGURE 1

Two balls of equal mass, one white and the other grey, are involved in a perfectly elastic collision. The grey ball is initially at rest and is struck by the white ball moving initially to the right at 5 m s^{-1} as shown in **FIGURE 1**. After the collision, the white ball moves in a direction that makes an angle of 32^{o} above the horizontal axis while the grey ball makes an angle of 51^{o} below the horizontal axis. Determine the speed of each balls after the collision.

[5 *marks*]

4. (a)

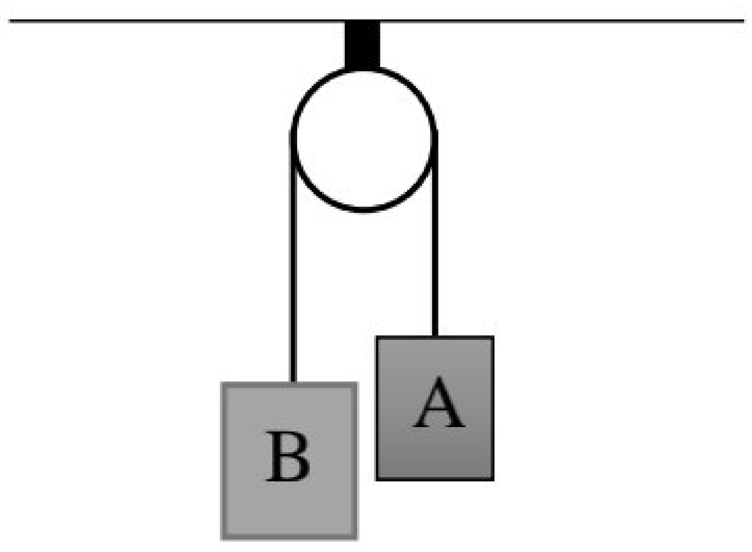

FIGURE 2

FIGURE 2 shows two boxes **A** and **B** hung by a light string through a frictionless pulley. The mass of box **B** is twice the mass of box **A**. Calculate the tension in the string and acceleration of boxes **A** and **B**, if the mass of box **A** is 250 g.

[5 *marks*]

(b)

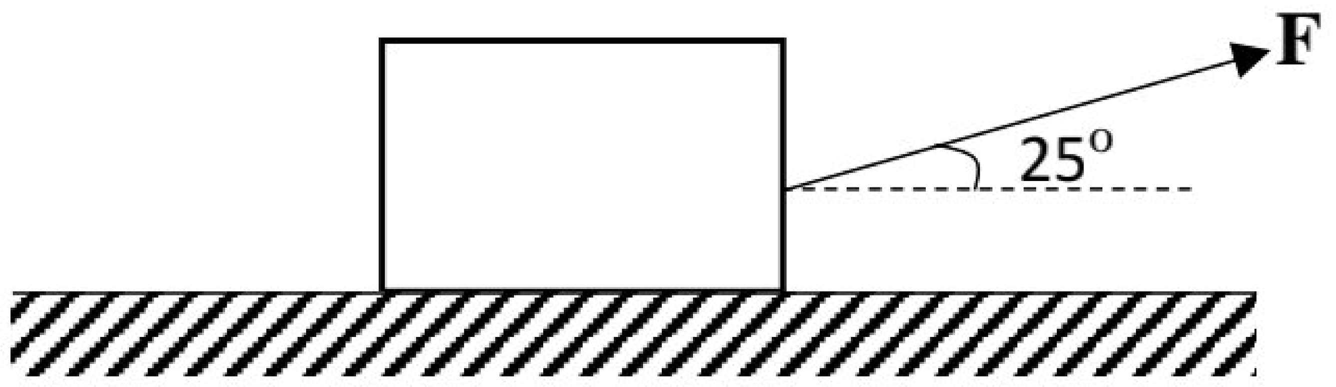

FIGURE 3

FIGURE 3 shows a 680 N box is pulled by a force, **F** at an angle of 25° to the horizontal. The box moves at a constant speed. The coefficient of kinetic friction is 0.50.

(i) Sketch the free body diagram on the box.

(ii) Calculate the reaction force and pulled force, **F** exerted on the box.

[7 *marks*]

5. (a) The pilot of a 60 500 kg jet plane is flying in circles whose diameter 1×10^5 m. It takes 3.6×10^3 s to make two rotation. Calculate

(i) the velocity and

(ii) the period of the plane.

[5 *marks*]

(b) A 50 kg child stands at the rim of a merry-go-round of radius 2.00 m, rotating with an angular speed of 3.00 rad s^{-1}.

(i) What is the child's centripetal acceleration?

(ii) What is the minimum force between her feet and the floor of the carousel that is required to keep her in the circular path?

[4 *marks*]

6. (a) A flywheel is speeded up from 5 r.p.m to 11 r.p.m in 100 s. The radius of the flywheel is 0.08 m. Calculate

(i) the angular acceleration of the flywheel.

(ii) the tangential acceleration of a point on the rim along its circular path.

(iii) the number of turns made by the flywheel.

[7 *marks*]

(b)

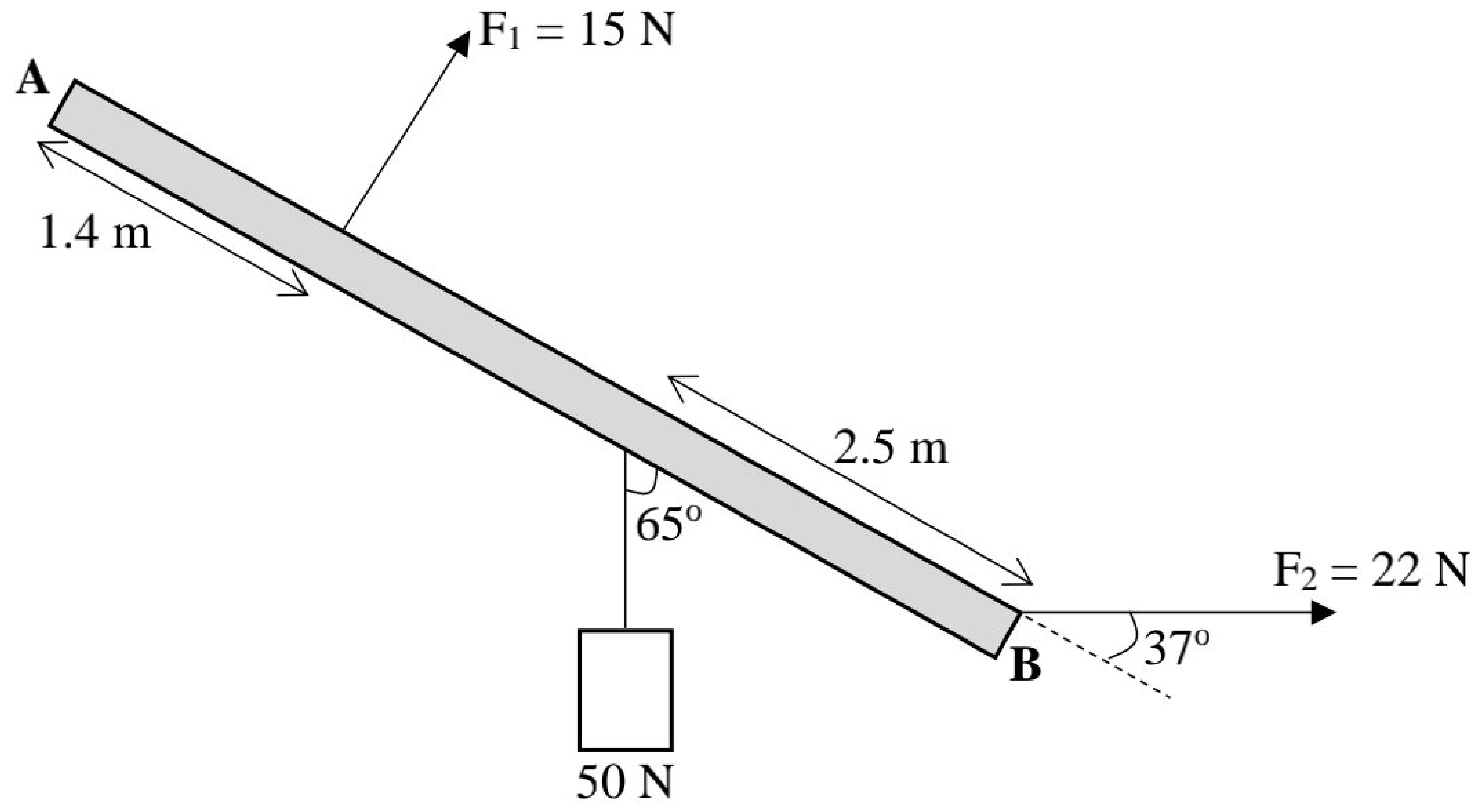

FIGURE 4

FIGURE 4 shows two forces F_1 and F_2 of 15 N and 22 N exerted on a uniform plank of 6 m. A hanging load having weight 0f 50 N is hung 2.5 m from point B. Calculate the

(i) total torques and direction of torque at the point **A**. Assume the plank is massless.

(ii) total torques and direction of torque at point **B** if the mass of the plank is 2 kg.

[5 *marks*]

7. (a) An iron plate with an area of 0.1 cm^2 and a thickness of 4×10^{-3} m has both ends maintained at temperature 373 K and 223 K respectively. The thermal conductivity of the iron is 80 W m^{-1} K^{-1}. Calculate

(i) the temperature gradient.

(ii) the heat lost through the iron plate in one day.

[3 marks]

(b) At what temperature of gaseous oxygen molecules where the rms speed of gaseous oxygen molecules equal to that of hydrogen molecules at 47 °C? Given the molar mass of hydrogen is 2 g mol^{-1} and molar mass of oxygen is 32 g mol^{-1}.

[3 *marks*]

(c) (i) A system releases 125 kJ of heat while 104 kJ of work is done on the system. Calculate the change in internal energy for the system.

(ii) Calculate the work done on or by the system that absorbs 260 kJ of heat and experiences an increase in internal energy by 157 kJ.

[4 *marks*]

(d) Two moles of helium gas are initially at a temperature of 27 ^{o}C, filling a volume of 0.02 m^3 at a pressure of 2×10^5 Pa. The helium gas undergoes an expansion at constant pressure until its volume doubles. Throughout this process, the system performs 4000 J of work and the internal energy increases by 7483 J. Subsequently, the gas undergoes isothermal compression until its volume returns to the initial value of 0.02 m^3 at a pressure of 4×10^5 Pa. During the isothermal compression process, 6916 J heat is released.

(i) Sketch a graph of P versus V for both processes.

(ii) Calculate the heat absorbed or released during the expansion process at constant pressure and specify whether it is absorbed or released.

(iii) Calculate the work done by the helium gas during the isothermal expansion process and indicate the work is done on the system or work done by the system.

[8 *marks*]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT